

ReviewLens

Product Review Sentiment Analysis
Dashboard Using Sentiment and
Aspect-Based Sentiment Analysis



Presentation Roadmap

- Problem and project goal
- Datasets and preprocessing
- System architecture and workflow
- Methods and techniques
- Model performance
- Working prototype and demo results
- Limitations and next steps



Problem, Goal, and Scope

- Manual review analysis becomes slow and inconsistent when review volume grows
- The goal is to transform uploaded review files into sentiment, aspect, and theme insights
- The system is designed for batch CSV and Excel analysis

Dataset Overview

- Training dataset: Amazon Product Reviews.csv
- Total Rows: 568,454
- Sample Data: 150,000
- Training Data: 120,000
- Testing Data: 30,000
- Key fields: rating, review text, summary, product information
- Rating-to-label mapping: 1 to 2 negative, 3 neutral, 4 to 5 positive

Preprocessing and Analysis Pipeline

Validate file type and read CSV or Excel with encoding fallback

Auto-detect review text, rating, date, product, and summary columns

Clean text by lowercasing, removing HTML, URLs, emails, punctuation, and extra spaces

Run sentiment prediction, ABSA, theme extraction, trends, summaries, and export generation

Tokenize text, remove stopwords while keeping negations, and lemmatize when possible

System Architecture and Runtime Workflow

Frontend: React dashboard with Upload, Dashboard, Single Predict, and Model Info views

Backend: Flask API with analysis endpoints, background jobs, status polling, exports, and single-review prediction

Storage model: local files only, with uploads, saved models, and generated exports

Async workflow: start analysis, poll progress, return final dashboard payload

Methods and Techniques Used

Overall sentiment classification uses TF-IDF features with Logistic Regression

Vectorizer supports up to 50,000 features with unigram and bigram terms

Class weighting is used to improve handling of neutral and negative classes

Aspect-based sentiment analysis is rule-based and covers quality, price, delivery, taste, service, appearance, and usability

Theme extraction uses TF-IDF keywords, frequent n-grams, praise and complaint buckets, and word-cloud frequencies

Additional analytics include monthly trends, product summaries, aspect themes, and aspect trends

Current Model Performance

- Model type: Logistic Regression + TF-IDF

> Seen in Model Info

- Interpretation: overall accuracy is strong, but macro metrics show that class balance still matters

Working Prototype and User Flow

- Upload and Analyze view supports CSV, XLSX, and XLS files
- Async job flow shows progress messages and percent completion
- Dashboard includes Overview, Aspects, Themes, and Trends sections
- Single Predict view returns sentiment, confidence, probabilities, and detected aspects
- Model Info view shows health status, metrics, per-class scores, and confusion matrix
- Exports include processed CSV, dashboard JSON, and aspect-focused CSV or JSON

Limitations and Next Steps

- There is no database nor persistent job queue
- ABSA is rule based which helps interpretability but limits flexibility
- The reviews table is not yet presentation ready, so it is currently hidden while further refinement is ongoing.
- Future improvements: persistent jobs, more aspect coverage, and stronger models.

This is me waiting to say "no update" during a team meeting



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